Nama : Christoper Hutagaol

NIM : 201331036

Kelas : 2B - TTE

TTD :

Jawaban

① Dik : $L = 39,6\ \mu H$

$C = 1\ nF$

$f_0 = 850\ kHz$

$Q = 100$

Amplitudo = $100\ \mu A$

$f_1 = 850\ kHz$

$f_2 = 900\ kHz$

Dit : G dan V ?

Jawab

- $G = \frac{\omega_0 C}{Q} = \frac{2\pi f \cdot C}{Q} = \frac{2\pi \cdot (850 \times 10^3)(1 \times 10^{-9})}{100} = \boxed{53,4\ \mu \mho //}$

$\rightarrow R = \frac{1}{G} = \frac{1}{53,4 \times 10^{-6}} = 18,72\ k\Omega$

- $V_1 = I_1 R$

$= (100 \times 10^{-6})(18,72 \times 10^3)$

$V_1 = \boxed{1,872\ V //}$

- $Y_2 = \sqrt{G^2 + \left(2\pi fC - \frac{1}{2\pi fL}\right)^2}$

$= \sqrt{(53,4 \times 10^{-6})^2 + \left(2\pi (900 \times 10^3)(10^{-9}) - \frac{1}{2\pi (900 \times 10^3)(39,6 \times 10^{-6})}\right)^2}$

$Y_2 = 1,18\ m\mho$

$\rightarrow V_2 = \frac{I_2}{Y_2} = \frac{100 \times 10^{-6}}{1,18 \times 10^{-3}} = \boxed{84,74\ mV //}$

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② Dik : $R_s = 50\ \Omega$

$R_L = 600\ \Omega$

$f = 75\ MHz$

Dit : Rangkaian Penyesuai impedansi tipe L ?

Jawab

- $Q_s = Q_p = \sqrt{\frac{R_p}{R_s} - 1} = \sqrt{\frac{600}{50} - 1} = \sqrt{11} = 3,31$

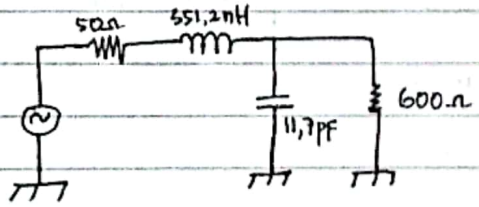
$X_s = Q_s R_s = (3,31)(50) = 165,5\ \Omega$

$X_p = \frac{R_p}{Q} = \frac{600}{3,31} = 181,26\ \Omega$ //

- $\omega L = 165,5\ \Omega \longrightarrow L = \frac{165.5}{2\pi (75 \times 10^6)} = 351,2\ nH$ //

- $\frac{1}{\omega C} = 181,26\ \Omega \longrightarrow C = \frac{1}{2\pi (75 \times 10^6) \cdot 181,26} = 11,7\ pF$ //

Rangkaian :



③ Dik : Penyesuai impedansi tipe T → 

$R_s = 50\ \Omega$

$R_l = 10\ \Omega$

$f = 100\ kH$

$Q = 10$

Dit : L ? dan C ?

Jawab

- $Q = \sqrt{\frac{R}{R_s} - 1} \rightarrow R = R_{kecil}\,(Q^2 + 1)$.

$R = 10\,(10^2 + 1) = 1010\ \Omega$ //

- $X_{s1} = Q R_s = (10)(50) = 500\ \Omega$ //

- $X_{p1} = \frac{R}{Q} = \frac{1010}{10} = 101\ \Omega$ //

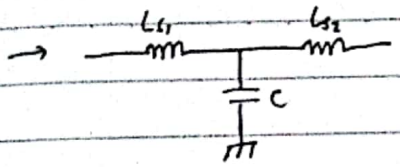
→ Q didefinisikan sbg dari R virtual dan R beban

- $Q = \sqrt{\frac{R}{R_L} - 1} = \sqrt{\frac{1010}{10} - 1} = 10$

- $X_{p2} = \frac{R}{Q_2} = \frac{1010}{10} = 101\ \Omega$ //

- $X_{s2} = Q_2 R_L = (10)(10) = 100\ \Omega$ //

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- $X_{p\,tot} = \frac{1}{X_{p_1}} + \frac{1}{X_{p_2}} = \frac{1}{101} + \frac{1}{101} = 50,5\ \Omega_{//}$

→ $C = \frac{1}{2\pi(100\times10^3)(50,5)} = \boxed{31,51\ nF_{//}}$

→ $L_{s_1} = \frac{101}{2\pi(100\times10^3)} = \boxed{160,7\ \mu H_{//}}$

→ $L_{s_2} = \frac{101}{2\pi(100\times10^3)} = \boxed{160,7\ \mu H_{//}}$

④ $Z_s = 25 + j15$

$Z_L = 100 - j25$

$N = 100$

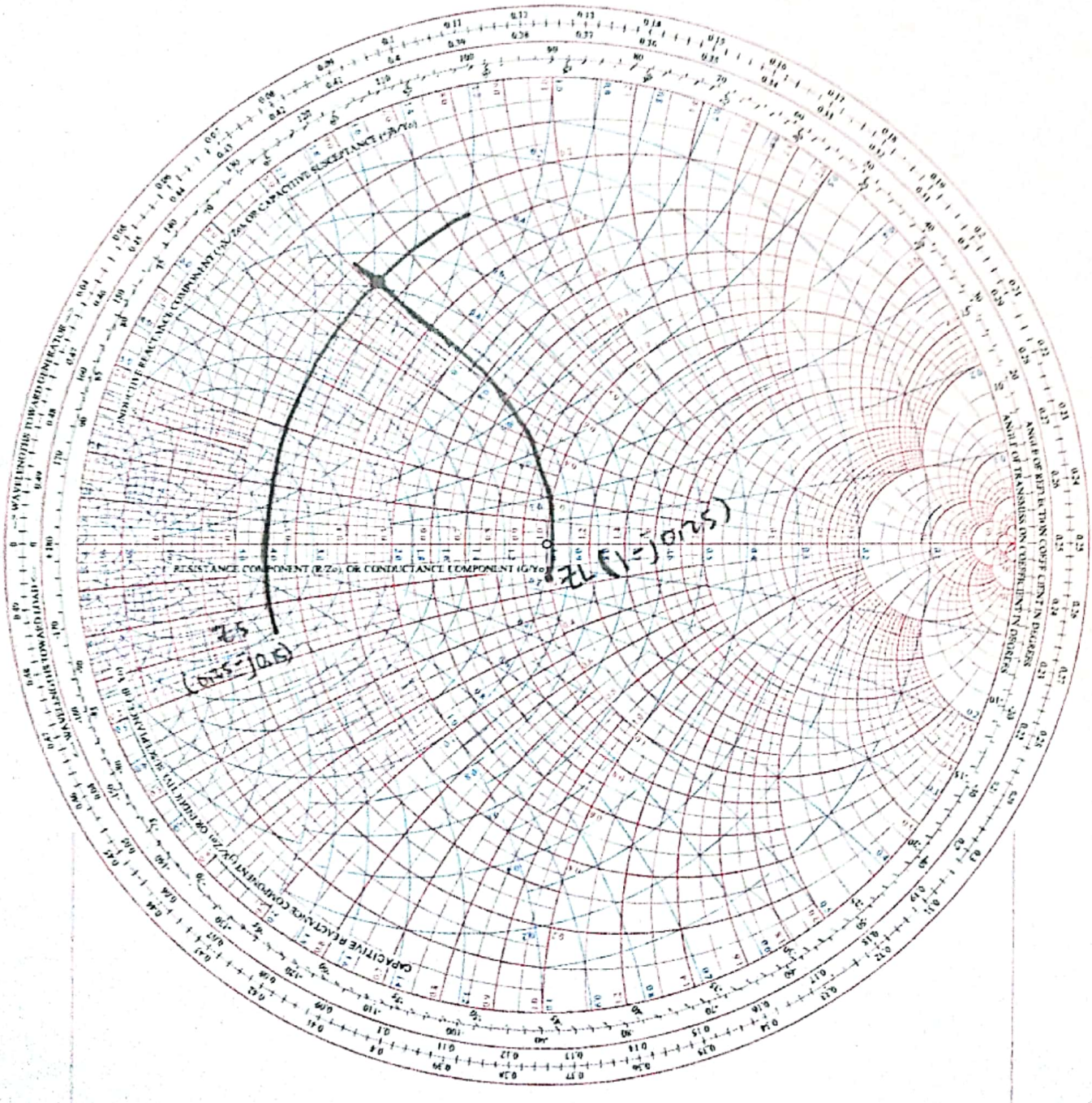
maka : $Z_s' = 0,25 + j0,15$

$Z_L' = 1 - j0,25.$

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NAME	TITLE	DWG NO.
Christoper Husaga	Jawaban UTS Nomor 4	
SMITH CHART	Courtesy of Microwaves101.com	DATE 25 Maret 2022

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES





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